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METHODOLOGIES AND APPLICATION

A novel machine learning method for estimating football players' value in the transfer market

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Abstract
Every year a huge amount of money is invested by the football clubs in the transfer window period to hire or release players. Estimating players' value in the transfer market is a crucial task for the managers of the clubs. Also, it is one of the attractive aspects of football for fans. Tranfermarkt.com is a reference website that determines the transfer fee of the players based on its members' opinions. The limitation of this website has attracted the attention of data scientists in recent years, resulting in creating datasets and data-driven estimating methods. In this paper, a novel method for estimating the value of players in the transfer market, based on the FIFA 20 dataset, is proposed. The proposed method has two phases. In the first phase, the dataset is clustered using an automatic clustering method called APSO-clustering. This automatic clustering method, which can detect the proper number of clusters, has divided the dataset into 4 clusters automatically indicating the position of the players: goalkeepers, midfielders, defenders, and strikers. In the second phase, a hybrid regression method which is a combination of particle swarm optimization (PSO) and support vector regression (SVR), is used to build a prediction model for each clusters' data points. In this hybrid method, PSO is used for feature selection and parameter tuning of SVR. The achieved results show that the proposed method can estimate the players' value with an accuracy of 74%. Comparing the performance of PSO with 3 other metaheuristics, the results demonstrated the superiority of PSO over GWO, IPO, and WOA.

Keywords Particle swarm optimization · APSO-clustering · Support vector regression · Market value · Football · FIFA 20

1 Introduction

Football (association football) is the most popular sport in the world. Millions of fans support their clubs during a season. Besides the attractiveness of the game itself, there are some other attractive issues for the fans. One of them is transferring the players during transfer windows. The fans need to know which player their club will sign a contract or how much money the club wants to spend in the transfer window period. Moreover, it is a crucial task for the managers of the club to sign good contracts with the new players. Sometimes they decide to spend huge money to hire a superstar player, and sometimes they prefer to spend low money but invest in young talented players to prepare them for the future of the club. In both cases, estimating the values of the players accurately in the transfer market is necessary. For example, in 2015 FC Liverpool paid 41 million euros to TSG Hoffenheim club to hire Roberto Firmino while TSG Hoffenheim bought him with only 4 million euros 4 years earlier.

Transfermarkt.com is a reference website that determines the values of the players in the transfer market using the crowd estimate method. In this method, the members of the website estimate the players' values and then special members, called mentors, determine the values based on the other members' estimations. This method has different limitations: (1) Inexperienced members can reduce the

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accuracy of estimation. (2) It is applicable for the famous players who are well known to the fans. But for other players, playing in minor leagues or small clubs, it does not provide an accurate estimation. So it does not support scouting players in minor leagues. (3) Since transfermarkt.com works based on the members' opinions, the players' values are not updated frequently while the poor or great performance of a player in an important match can decrease or increase his/her value. So it is very necessary to have an accurate method to estimate players' values. Several types of research have been conducted in this field up to now that most of them are focused on analyzing the effect of different parameters (such as players' popularity based on the social media) on the players' value. On the other hand, in most of the researches, the players' position in the field and their special skills are not considered in the proposed models. This is due to the lack of a structured dataset presenting the players' skills in numerical format. Therefore, the main drawbacks of the existing methods, which are analyzed with more details in the next section, are lacking a structured dataset and not paying attention to the special skills of the players which makes them less applicable.

In this paper, a new effective machine learning method is introduced, designed using FIFA 20 dataset. This dataset includes different performance ratings of 18,278 players. FIFA 20 is a football simulation video game developed by Electronic Arts and released in 2020. Attributes of this dataset show different skills of players such as defending score, attacking score, and passing score; using this dataset we can evaluate the performance of the players in the past season. Definitely, better players have higher values in the transfer market. The proposed method which has shown a great performance in different experiments is trained in two phases. In the first phase, the players are divided into different clusters based on their skills using an automatic clustering method. Then in the second phase, for each cluster, a regression model is trained using the particle swarm optimization algorithm and support vector regression method. The main novelties of this research are:

1. Using a novel automatic clustering method in the first phase to divide the data points (players) into different groups. This automatic clustering method has put the players with the same roles in the same clusters.
2. Training a hybrid regression method for each cluster. In this hybrid method, PSO has been used for feature selection and tuning the parameters of SVR. Feature selection for each cluster has shown the effect of different skills of different players on their market value and parameter tuning has increased the accuracy of SVR in learning the hidden patterns.

The main challenges of this research are complexity and the huge amount of data. Clustering the data points in the first phase reduces the amount of data that should be learned by each SVR in the second phase. In other words, learning the hidden patterns in each cluster by an independent SVR model is an easier task in comparison with learning the whole data by a single regression method since each cluster has similar data points and also the amount of data in each cluster is much less than the whole data. Furthermore, eliminating redundant features reduces the complexity and ambiguity of the data which results in better accuracy. The paper is organized in the following manner: in the next section, the recently related works are presented and analyzed. In the third section the dataset, used in this research, is completely described. The proposed method is explained completely in Sect. 4. Finally, Sects. 5 and 6 are devoted to experimental results and conclusion, respectively.

2 Related works

Müller et al. (2017) introduced a multi-level regression method for estimation of players' market value. They created a dataset containing different attributes such as player characteristics (age, position, and nationality), player performance, and popularity. They have analyzed the effect of different parameters on the players' market value, and then they have trained a regression method for value estimation. Also, in their paper authors explained the limitations of the crowdsource estimating method used by transfermarkt.com. Their method has shown good accuracy for low- to medium-priced players, while for high-priced players crowd has shown better performance. In fact, their method has performed well for the lower 90% of the transfers, while for the top 10% of the players the model estimations were inaccurate. This is probably because of lacking important intangible indicators such as player's potential to boost tickets of jerseys sales. On the other hand, professional football clubs sometimes pay very high transfer fees for superstars which may not reflect their true values. In another research (Herm et al. 2014), a method based on the players' talent and judgment variables is introduced to estimate the transfer fee of football players. In this project, the influence of different player attributes on their market value, which are proposed by transfermarkt.de, is tested. They have focused on 338 active players who played in Bundesliga during the first half of the season 2011–2012. The authors indicated that the community's estimations can be largely predicted by using an econometric model containing two blocks: talent-related measures and variables that result from judgments of external experts such as coaches and journalists. Focusing

on 5 five talent variables (age, precision, success, assertion, and flexibility), their analysis showed that only age has a nonlinear relationship with the market value of the players. In this research, the values estimated by the fans are used to develop a model. So the main drawback of this research is using community evaluations that can be biased or suffered from a lack of knowledge. Franck and Nüesch (2012) investigated the effect of talent and popularity of the players in their market value. They tried to measure the talent of the players using 20 quantifying parameters in correlation with their contribution toward the team's success. Using an OLS regression model, they have concluded that both talent and non-performance-related popularity of the players increase their market value. Kiefer in a research project (2012) analyzed the effect of players' performance on their popularity and market value. In her research, the popularity of the players and their market value are measured 1 day before the Euro 2012 tournament and also 1 day after the tournament. The popularity is measured using three major websites: google.com, facebook.com, and Uefa.com, and for the market value, transfermarkt.com has been used. According to this study, showing a good performance in a big national tournament, like Euro 2012, can increase the popularity of the players significantly and as a result lead to an increase in the market value. Majewski (2016) investigated the effect of different factors on the value of forward players in order to determine the most important factors. In this study, he used the information of 150 famous strikers stored in transfemarkt.de and adopted a GLS (generalized least squared) method to find the important factors. Based on his results, the number of goals, assists, the value of the whole team, and FIFA ranking points had an impact on the market value of the forward players. Considering the role of the players is the positive point of his study, but just focusing on the forward players is a negative point. Also, lacking a structured dataset, including the information of players in numerical format, is the main drawback.

Singh and Lamba (2019) used FIFA 18 game rating data in addition to the data used in the previous works to estimate the players' values. The dataset included popularity indicators (Wikipedia views on a player's article), FIFA game ratings, consistency indicators (player's talent and football skills), and previously known market value of the player. The main goal of their research was to find the influence of different attributes of the used dataset on the market value of the players. To make a good estimation model, after selecting the most important features using correlation, the performance of four different machine learning methods is analyzed. Yigit et al. (2019) have proposed a method based on a dataset from a football manager simulation game to estimate the value of players in the transfer market. Football manager is a simulation game developed by Sports Interactive and published by Sega. The data contains 49 attributes, demonstrating different skills of players, under 4 main categories: technical, mental, physical, and goalkeeping. Each attribute is a number between 0 and 20 which is determined by considering the opinion of professional scouts. The value of each player is added from the transfermarkt.com website. Different regression methods have been analyzed to find the best set of predictors. The results showed that only age has a significant nonlinear relationship with the target value. To compare the performance of different regression models, mean squared error (MSE) has been calculated in this research. Also, the effect of different attributes on the values of players with different positions is investigated. In a research conducted recently, Felipe et al. (2020) analyzed the effect of team variables and player positions on the market value of football players. In their study, all of the players in the top 5 European leagues (Spain, England, Italy, France, and Germany) from the 2017 to 2018 season were included. According to their results, regression analysis showed that the level of the team, the birth month, the league, the playing position, and the age of the player affect the players' market values. Also, they indicated that the players participating in the Uefa Champions League of the Premier League, playing as attacking midfielders and born in the first quarter of the year are the most valued players.

In most of the researches conducted in this field, the regression models have been used without paying attention to the fact that different players with different positions have different skills. For example, to estimate the value of a goalkeeper, goalkeeping skills should be considered, while for a forward player the attacking skills such as speed, heading, and dribbling should be considered. Although in some researches the effect of different attributes (skills) of the dataset has been examined using correlation, still having an automatic method capable of finding the relevant attributes of players based on their position is missed. On the other hand, in the most of these researches the main goal was to investigate the effect of different factors on the players' market value while in this research, a new and novel method based on automatic clustering and optimal support vector regression is introduced which can automatically detect the important factor for different groups of players.

Optimization methods, especially metaheuristic optimization algorithms, are powerful methods that can help the researchers to overcome the difficulties and challenges in different fields. Several types of research have been done recently in different fields in which optimization algorithms have played key roles. For example, Gadekallu et al. (2020) have introduced a feature selection method based on principle component analysis (PCA) and whale

optimization algorithm (WOA). In their method, PCA is used for dimensionality reduction, and WOA is used to find the best feature subset. They have used their method for the classification of tomato plant diseases. In another research, Swarna Priya et al. (2020) have proposed an Energy-Efficient Cloud-Based Internet of Everything (EECloudIOE) architecture. In this new architecture, wind-driven optimization and firefly optimization algorithms are used to optimize the energy utilization in the IoT network. Deepa et al. (2020) have used an optimized classifier, called RASGD (Ridge-Adaline Stochastic Gradient Descent) for earlier prediction of diabetes mellitus. In their research, to minimize the cost function of the classifier, the RASGD adopts an unconstrained optimization model. Also to increase the convergence speed of the classifier, the Adaline Stochastic Gradient Descent classifier is integrated with ridge regression. Maddikunta et al. (2020) have used a hybrid optimization algorithm for green communication in IoT networks. In their research, the proposed method, which is designed by a combination of whale optimization and moth-flame optimization algorithms, is used to optimize the energy utilization of network that can be affected by several factors such as cluster head (CH), number of alive nodes, residual energy, and temperature.

In recent years, several metaheuristics algorithms are introduced. In this research, PSO is used due to its low complexity in comparison with other metaheuristics and also effectiveness in solving hard and complex optimization problems which are addressed in several kinds of literature (Zhang et al. 2020; Liang et al. 2020; Khosravy et al. 2017).

3 Data description

FIFA 20 dataset, containing the characteristics of 18,278 players, is used to build a prediction model. FIFA 20 is a football simulation video game published by Electronic Arts as part of the FIFA series. The dataset includes players' attributes with statistics such as attacking skills, defense mentality, and GK skills. Also, the market value of the players in the career mode of the video game is included in the dataset as well as players' data like age, height, preferred foot, club, and national reputation. Briefly, the dataset has 18,278 objects (players) and 55 attributes. In Table 1, some of the features of the players and their possible values are mentioned. In Table 2, four rows of the dataset containing some of the characteristics of, Lionel Messi, Cristiano Ronaldo, Jan Oblak, and Virgil van Dijk are shown. According to Table 2, Atletico Madrid's goalkeeper, Jan Oblak, while some of his skills are set to zero, is more expensive than Cristiano Ronaldo, one of the greatest strikers in football. This means that the players' market value should be estimated based on their positions in the pitch. Furthermore, to have a better accuracy different subsets of features for different players based on their positions should be selected. To reach this goal, an automatic clustering method, which we have introduced in our previous works (Behravan et al. 2018, 2019) is used to divide players into different groups (clusters) based on their skills. After that, for each group an optimal support vector regression method is trained separately. In the next section, the proposed method is explained thoroughly.

4 Proposed method

To estimate the market value of players, a two-phase method is presented in this paper. In the first phase, the dataset is grouped into different clusters using an automatic clustering algorithm called APSO-clustering. APSO-clustering, which is designed using the PSO algorithm, searches the solution space to find the proper number of clusters and the position of the centroids simultaneously. The effectiveness and the power of this method are proved in our previous studies. In the second phase, for each cluster, a hybrid regression method called PSO-SVR is trained. In the next subsections, PSO, APSO-clustering method, and PSO-SVR are completely clarified.

4.1 Particle Swarm Optimization

PSO is a metaheuristic optimization algorithm suggested by Kennedy and Eberhart (1995). This effective optimization algorithm is created by mathematical modeling of social behaviors of different species such as birds' flocking and fishes' schooling. Finding the optimum point of an objective function, PSO uses a population of individuals, called particles. These particles explore and exploit different parts of the solution space by cooperating during the search process. At the beginning of the process, particles are randomly generated. Then, their fitness value is calculated using the objective function. After appointing the best particle as the leader, the particles start to move in the solution space. The following equations show how particles change their position.

$$v^{t+1} = w \cdot v^t + C_1 \cdot \text{rand} \cdot (p_{\text{best}} - x^t) + C_2 \cdot \text{rand} \cdot (p_{\text{gbest}} - x^t) \quad (1)$$

$$x^{t+1} = x^t + v^{t+1} \quad (2)$$

In Table 3, the definition of the parameters used in Eqs. 1 and 2 is mentioned. After the movement of the particles, their fitness value is updated followed by updating the position of the leader. Finally, after a predefined number of

Table 1 Some of the attributes of the dataset

Features	Possible value	Description
Age	–	–
Height	–	–
Weight	–	–
Preferred foot	1 or 2	Right = 1, Left = 2
International reputation	1, 2, 3, 4 or 5	The reputation of the national team of the player based on FIFA world ranking
Passing	1–100	The passing ability of the players
Shooting	1–100	The accuracy and the power of shoots. Important for the strikers
Pace	1–100	How fast the player run
Dribbling	1–100	Dribbling skill of the player.
Attacking_heading_accuracy	1–100	The heading accuracy of the strikers
Defending	1–100	The defending skill of the defenders and defending midfielders.
Gk_diving	1–100	The diving ability of goalkeepers.
Market value	0.04–95.5	Player's value in million euro

Table 2 Some of the characteristics of four players

Player	Preferred foot	International reputation	Passing	Shooting	Pace	Dribbling	Heading accuracy	Defending	Gk_diving	Market value
L. Messi	2	5	92	92	87	96	70	39	0	95.5
C. Ronaldo	1	5	82	93	90	89	89	35	0	58.5
J. Oblak	1	3	0	0	0	0	15	0	87	77.5
V. van Dijk	1	3	70	60	77	71	86	90	0	78

Table 3 Definition of the parameters used in Eqs. 1 and 2

Parameter	Description
V^{t+1}	Particle's velocity in iteration $t + 1$
w	Inertia coefficient
C_1	Cognitive factor
C_2	Social factor
X^t	Particle's position in iteration t
P_{best}	The best position of the particle from the beginning of the process
P_{gbest}	The position of the leader
rand	Random number with normal distribution between 0 and 1

iterations, the position of the leader is presented as the output of the algorithm. In Fig. 1, the flowchart of PSO is shown.

4.2 APSO-clustering method

APSO-clustering is an automatic (in some works called unsupervised) clustering method that can find the proper number of clusters in addition to the position of the centroids. The main merit of this method is its high accuracy in partitioning complex and big datasets in comparison with other automatic clustering methods such as X-means (Pelleg and Moore 2000). This novel clustering method works on two levels. In the first level, the main task is to find a suitable number of clusters, and in the second level, detecting the exact position of the centroids is the main task. In both levels, the PSO-Clustering method is the basic block. This basic block searches the solution space to

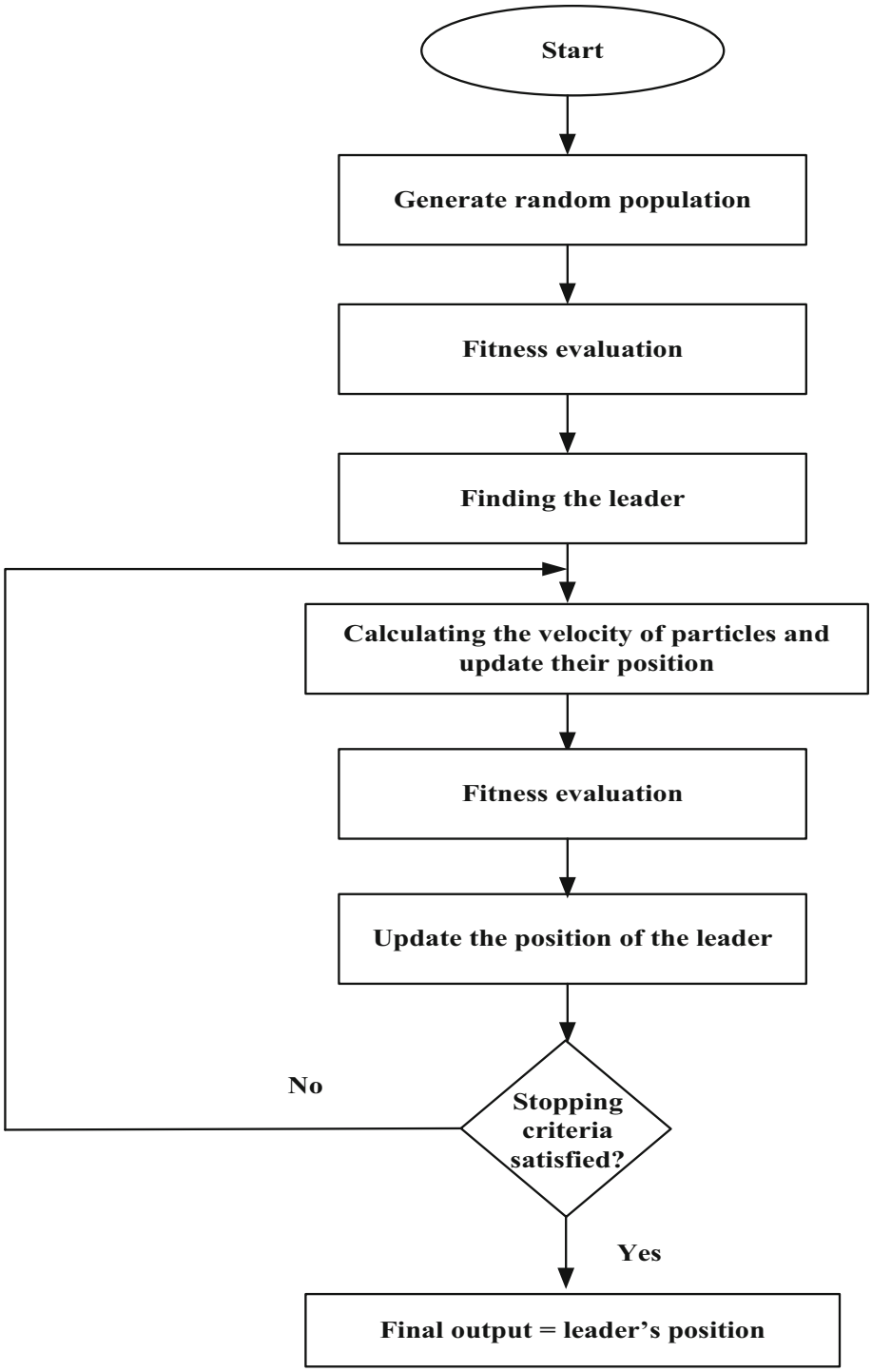


Fig. 1 Flowchart of PSO

find k centroids, while k is predetermined by the user. Therefore, particles contain $k \times p$ cells where p is the number of features in the dataset and k is the number of clusters. Also, the Calinski–Harabasz index (Maulik and Bandyopadhyay 2002) is used for fitness evaluation. In the first level of the APSO-clustering method, a sequence of the basic block is run to detect the proper number of clusters. The sequence consists of 10 steps, and in different steps, PSO-Clustering is run with different values of k. At the end of the first level, the best number of clusters is detected based on the fitness values of the best solutions found in different steps of the sequence. In the second phase, the PSO-Clustering method searches the solution space to find k centroids, while k is the output of the first level. In the second level, the number of the particles and the iteration number are set to 50 and 600, respectively, to

find the exact position of the centroids. These numbers are 5 and 150 in each step of the sequence in the first level. The pseudocode of the APSO-clustering method is demonstrated in Fig. 2.

4.3 PSO-SVR

Generally, feature selection is an important preprocessing technique that can reduce the complexity of the data and improve the accuracy of the machine learning method. Several kinds of research have been conducted in the field of feature selection introducing different methods such as Abualigah et al. (2018), Abualigah (2019). In our previous research, we introduced a hybrid classification method based on multi-objective PSO (MOPSO) and support vector machine called MOPSO-SVM (Behravan et al. 2016). MOPSO has been used to find the proper kernel function, tune the kernel parameter (for RBF and polynomial kernel functions), find the proper value of penalty factor, and select the best subset of features for two objective functions: recognition score and reliability. In this research, a similar method is used for support vector regression (Drucker et al. 1997). Since players have different skills based on their position in the pitch, their market value should be estimated based on their position and corresponding skills. So feature selection is of great importance in this research. Therefore, PSO is used to find the best subset of features and detect the proper value of the Gaussian kernel. In fact, during the training process, PSO selects a subset of features and a value for the Gaussian kernel's parameter. Then, SVR is trained with the Gaussian kernel and its corresponding parameter value and also training data containing the selected features.

Selecting the best features among 55 features of the dataset is a hard task due to the complexity of the solution space. Each search agent is a vector with the size of 56 in which 55 cells are used for feature selection and 1 cell is used for detecting the value of γ. For feature selection, a binary encoding scheme is used which means that the amount of each cell is 1, for selecting, or 0 for eliminating the corresponding feature. Therefore, the best feature subset should be found among 2^{55} subsets. It is an NP-hard optimization problem that can be solved by metaheuristic algorithms. Mean squared error (MSE) function is used for the fitness evaluation. The fitness amount of each particle is calculated using a threefold cross-validation method. The flowchart of the PSO-SVR is shown in Fig. 3.

5 Experimental results

As mentioned in the previous section, in the first phase of the proposed method the dataset has been partitioned using APSO-clustering. This method has partitioned the samples (players) of the dataset into 4 clusters. Based on the output of this phase, the players can be divided into 4 groups

Fig. 2 Pseudocode of the APSO-clustering method

```
Stage 1:
Inputs: number of sequences, length of sequence, number of population
For i=1 to number of sequences do:
        1.k = Generate a random integer number
        2. beginning population = Generate a random population
        3.Current state.fitness = inf
        4. Current state.k=Ø
        for j=1 to length of sequence do:
            if (j==1) do:
                current state.k=k
            end
            best fitness = PSO-Clustering(k, beginning population)
            if best fitness < current state.fitness
                current state.fitness=best fitness
                curent state.k=k
            end
            k = current state.k ± ε
        end
Bestk[i] = current state.k
End
Output: find the best solution which has the best fitness amount and its corresponding k
and its corresponding beginning population
Stage 2:
Final output = PSO-Clustering(best k, beginning population)
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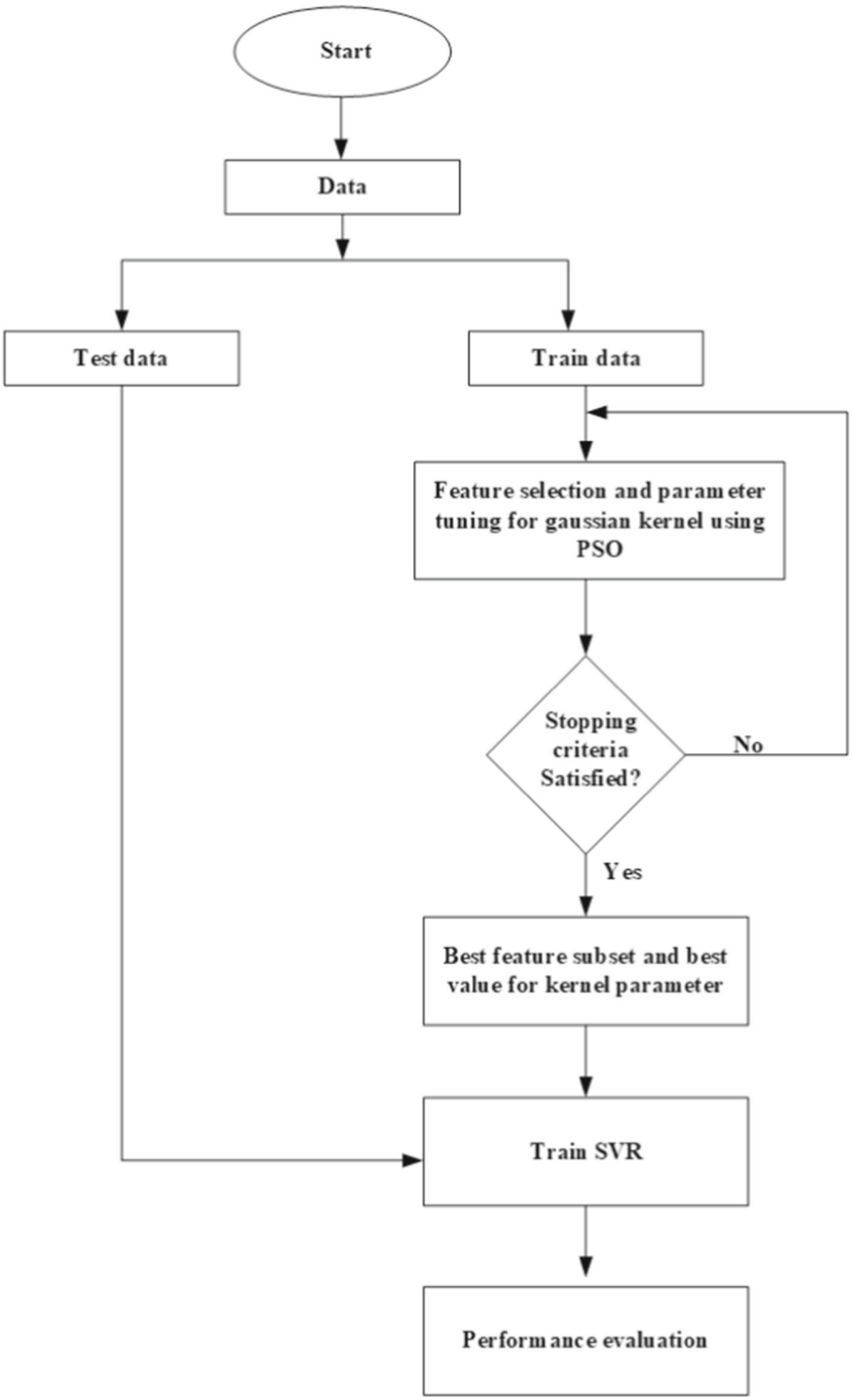


Fig. 3 Flowchart of PSO-SVR

depending on their skills. These groups show different roles in football including goalkeepers, defenders, midfielders, and strikers. The results achieved for each of these clusters are thoroughly presented in the next subsections.

5.1 First cluster: goalkeepers

The first cluster detected by the APSO-clustering method contains the goalkeepers. Some of the famous goalkeepers, their club, nationality, and some of their corresponding features are shown in Table 4.

Table 4 Top 6 goalkeepers included in the first cluster and some of their features in the FIFA 20 dataset

Player	Club	Nationality	overall	Gk_diving	Gk_handling	Gk_reflexes	Gk_positioning	Market value
M. ter Stegen	Barcelona	Germany	90	88	85	90	88	67.5
Allisson	Liverpool	Brazil	89	85	84	89	90	58
De Gea	Manchester United	Spain	89	90	84	92	85	56
T. Courtois	Real Madrid	Belgium	88	85	88	87	85	48
S. Handanovic	Inter Milan	Slovenia	88	88	85	89	89	26
M. Neuer	Bayern Munchen	Germany	88	87	87	87	85	32

This cluster has 1404 samples which means that 1404 goalkeepers are included in this dataset. In the second phase of the proposed method, a PSO-SVR method is trained on this cluster. In this regression method, PSO focused on finding the best feature subset or in another point of view eliminating redundant features in addition to finding the optimal value for the Gaussian kernel's parameter. The best value found for the Gaussian kernel's parameter is 65, and 29 features are selected for this cluster that some of them are indicated in Table 5. Table 5 shows the most important skills in determining the goalkeepers' market value. According to this table, it can be understood that, besides their special skills such as *Gk_handling* and *Gk_speed*, there are some other general skills like *ball control* and *shot power* which are important in goalkeepers' market value estimation, as well as personal features like *weight*. The most interesting point in this table is that *age* is excluded as an effective factor in goalkeepers' market value, while it is very important for other players. Furthermore, according to this table, *height* and *international reputation* do not influence the goalkeepers' market value.

5.2 Second cluster: strikers

APSO-clustering method has put strikers (or forwards) in the second cluster. Some of the well-known strikers in this cluster and their corresponding features are shown in Table 6. According to Table 6, the market value of Manchester City's striker, Raheem Sterling, is more than Cristiano Ronaldo. This is also true for K. Mbappe, Neymar Jr, and Hazard, while Ronaldo has a better *overall* value than all of these players. This point indicates that for this group of players besides special skills another kind of factor (like *age*) has a great effect on the market value. In the second phase, PSO-SVR has selected 32 features for this cluster. Also, the best value for the kernel's parameter is 103. Some of the selected features for this cluster are presented in Table 7. According to Table 7, it is clear that *age*, *height,* and also *weight* of the strikers are important in their market value, while for the goalkeepers just *weight* is selected. Also, another important skill for the strikers is *weak_foot* which shows their ability in using their weak foot. Definitely, the strikers need to be able to use their weak foot in some situations as most famous strikers, like Cristiano Ronaldo and Barcelona's former striker Ronaldinho, can shot or pass the ball with both right and left foot accurately.

Table 5 Some of the selected features for the first cluster

Feature number	Feature name
3	Weight
4	Overall
15	Physic
17	Gk_handling
19	Gk_reflexes
20	Gk_speed
31	Ball control
38	Shot power

5.3 Third cluster: defenders

Some of the famous defenders among 3464 players included in this cluster are shown in Table 8. Thirty features are selected for this cluster, of which eleven features are presented in Table 9. Also, the best value for kernel's parameter is 57. According to Tables 8 and 9, *Physic* is an important factor for the defenders' value in the market. Besides that, *age* and *height* are also important. Also, Table 9 indicates that another important factor in defenders' market value is *shooting* skill, while Table 8 shows that best defenders have *shooting* skill less than 50. These numbers show the complexity of determining the important factor of the players' market value estimation.

5.4 Fourth cluster: midfielders

This cluster has 3801 midfielder players. Some of the well-known midfielders included in this cluster are shown in Table 10. PSO has selected 28 features for this cluster;

Table 6 Top 6 strikers included in the second cluster and some of their features in the FIFA 20 dataset

Player	Club	Nationality	overall	Shooting	Crossing	Attacking finishing	Heading accuracy	Market value
L. Messi	Barcelona	Argentina	94	92	88	95	70	95.5
C. Ronaldo	Juventus	Portugal	93	93	84	94	89	58.5
Neymar Jr	Paris Saint-Germain	Brazil	92	85	87	87	62	105.5
E. Hazard	Real Madrid	Belgium	91	83	81	84	61	90
K. Mbappe	Paris Saint-Germain	France	89	84	78	89	77	93.5
R. Sterling	Manchester City	England	88	79	78	83	38	73

Table 7 Some of the selected features for the second cluster

Feature number	Feature name
1	Age
2	Height
3	Weight
4	Overall
6	Preferred foot
8	Weak foot
10	Pace
11	Shooting
28	Skill_curve
30	Long_passing
31	Ball control

Table 9 Some of the selected features for the third cluster

Feature number	Feature name
1	Age
2	Height
4	Overall
7	International reputation
8	Weak foot
11	Shooting
15	Physic
30	Long_passing
31	Ball control
38	Power_jumping
48	Defending marking

some of them are demonstrated in Table 11. Also, it should be mentioned that the best value for kernel's parameter detected by PSO is 76. In this cluster, like the third cluster, *age* and *height* are among the selected features. Also, *international reputation* is important for the estimation of the midfielders' market value. Depending on the position of the midfielders in the pitch, there are different kinds of midfielders in the lineup of a team such as *central midfielder* (CM), *defending midfielder* (DM), *attacking midfielder* (AM), *left and right midfielders* (LM and RM). So it is reasonable to have both defending and attacking skills in the selected features. According to Table 11, another important feature for these players is *free_kick_accuracy* which is not selected for the other clusters.

5.5 Estimating test players' value

For estimating test players' value, each player's cluster should be distinguished at the first step. In another point of view, each test sample should be assigned to the closest centroid. Then in the second step, the value of the player is estimated using the regression model trained on the

Table 8 Top 6 defenders included in the third cluster and some of their features in FIFA 20 dataset

Player	Club	Nationality	overall	Shooting	Defending	Physic	Movement reaction	Market value
K. Koulibaly	SSC Napoli	Senegal	89	28	89	87	86	67.5
M. Skriniar	Inter Milan	Slovakia	86	41	87	82	83	52
K. Manolas	SSC Napoli	Greece	85	25	86	81	82	37.5
Felipe	Atletico Madrid	Brazil	84	41	84	87	80	25
A. Romagnoli	AC Milan	Italy	83	40	85	79	78	32
J. Tah	Bayer 04 Leverkusen	Germany	83	27	83	81	81	31

Table 10 Top 6 midfielders included in the fourth cluster and some of their features in the FIFA 20 dataset

Player	Club	Nationality	overall	Passing	Dribbling	Defending	Attacking-short-passing	Market value
K. De Bruyne	Manchester City	Belgium	91	92	86	61	92	90
L. Modric	Real Madrid	Croatia	90	89	89	72	92	45
N. Kante	Chelsea	France	89	77	81	87	86	66
S. Busquets	Barcelona	Spain	89	80	80	85	89	55
P. Pogba	Manchester United	France	88	86	85	66	86	72.5
C. Eriksen	Inter Milan	Denmark	88	90	84	53	91	68

Table 11 Some of the selected features for the fourth cluster

Feature number	Feature name
1	Age
2	Height
4	Overall
7	International reputation
10	Pace
12	Passing
13	Dribbling
29	Free kick accuracy
31	Ball control
38	Power_jumping
48	Defending marking

corresponding cluster's data and the features selected by the PSO in the training phase. In Table 12, some of the test players, their corresponding cluster, and their estimated and real values are shown.

5.6 Performance evaluation

The second phase of the proposed method is implemented with three other metaheuristics, including GWO (Mirjalili et al. 2014), IPO (Mozaffari et al. 2016), and WOA (Mirjalili and Lewis 2016) for comparison purposes. Therefore, the accuracy of four methods (PSO-SVR, GWO-SVR, IPO-SVR, and WOA-SVR) is measured through 5 different evaluation metrics (MSE, RMSE, MAE, MAPE, and R^2) and shown in Table 13.

Table 13 Accuracy measurements of PSO-SVR, GWO-SVR, IPO-SVR, and WOA-SVR

Method	MSE	RMSE	MAE	MAPE	R^2
PSO-SVR	7.94×10^{12}	2.81×10^{6}	7.11×10^{5}	42.26	0.74
GWO-SVR	9.25×10^{12}	3.04×10^{6}	8.31×10^{5}	45.30	0.70
IPO-SVR	1.02×10^{13}	3.19×10^{6}	9.45×10^{5}	59.83	0.67
WOA-SVR	1.11×10^{13}	3.33×10^{6}	9.69×10^{5}	62.99	0.64

According to this table, PSO-SVR has shown better performance in comparison with other methods. The R^2 index for PSO-SVR is 0.74 which means that the accuracy of PSO-SVR is almost 74%. In other words, PSO-SVR can estimate the value of players with 74% accuracy. The accuracies achieved by GWO, IPO, and WOA are 70%, 67%, and 64%, respectively. Other evaluation metrics, which indicate the error rates, confirm the superiority of PSO over GWO, IPO, and WOA. In Table 14, the details of the results achieved by these four methods are presented.

According to Table 14, PSO has selected 29, 32, 30, and 28 features from the first, second, third, and fourth clusters, respectively. Also, the values detected, by PSO, for γ, are 49, 45, 63, and 45, respectively. Although, run time is not

Table 12 Some of the test players and their estimated values

Player	Club	Cluster	Estimated value	Real value
Marcelo	Real Madrid	4	24,804,697	28,000,000
Lucas Leiva	SS Lazio	4	18,903,401	20,500,000
A. Vidal	Barcelona	4	21,233,354	23,500,000
M. Gotze	Borussia dortmund	4	23,715,814	26,000,000
Willian	Chelsea	4	17,825,667	21,000,000
G. Buffon	Juventus	1	6,346,677	2,300,000
Falcao	Galatasaray SK	2	11,799,271	16,500,000
E. Can	Borussia dortmund	4	21,480,327	23,000,000

Table 14 The details of the results achieved by four methods

Method	Number of selected features for each cluster	Values of γ achieved for each SVR	Run time (s)
PSO-SVR	29, 32, 30, 28	65, 103, 57, 76	18,925
GWO-SVR	24, 19, 17, 16	61, 28, 50, 79	13,688.25
IPO-SVR	30, 24, 10, 25	34, 138, 423, 69	23,654.78
WOA-SVR	11, 22, 27, 44	358, 121, 50, 92	17,694.37

an important issue in this problem; in the last column of Table 14, the run times of the methods are shown. According to these numbers, GWO and PSO are the first and third algorithms in terms of run time, respectively. Briefly, Tables 13 and 14 show the effectiveness of PSO in solving hard and complex optimization problems. Moreover, another interesting method introduced for players' market value estimation (Müller et al. 2017) is selected for comparison. According to the authors, the RMSE and MAE for their method are 5,793,474 and 3,241,733, respectively, while these numbers for our method (PSO-SVR) are 2,819,286 and 711,029.413. These numbers indicate that our method has great superiority over the other methods in the field of football players' market value estimation.

6 Conclusion

To estimate the players' market value, it is necessary to consider their special skills based on their position in the pitch. For example, the skills which are important for the goalkeepers should not be considered for the strikers. Sometimes it is very hard to extract relevant features of the players manually from a structured dataset. Definitely, this task should be done considering the position of the players. In this research, an effective method for predicting the football players' value in the transfer market is introduced based on FIFA 20 dataset. In the training phase, the dataset is clustered using the APSO-clustering algorithm which resulted in detecting four clusters: goalkeepers, strikers, defenders, and midfielders. Then, for each cluster, an automatic regression method, able to detect the relevant features, is trained. In the testing phase first, the cluster of the test sample is distinguished; after that, the corresponding regression model estimates the market value of the test player. The main novelty of this research is using an effective automatic clustering method in the training phase to extract different groups of players and also estimating the players' value based on their relevant features. Using a heuristic feature selection method, which eliminates the redundant features based on the accuracy of the regression model, is a novel and innovative solution to this problem. The evaluation metrics, presented in Table 13, show the superiority of the proposed method over other introduced methodologies.

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Authors' contribution The main idea is first suggested by Dr. Seyed Mohammad Razavi, and it is developed and implemented by Iman Behravan (Corresponding author). Also, Dr. Razavi has contributed to the interpretation of the results. All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Availability of data and material The dataset used in this research (FIFA 20) is available on the Internet. Everyone can easily access the data.

Code availability All the simulations are done in MATLAB R2019b which has several toolboxes including all of the algorithms used in this research specially SVR and PSO.

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